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Human Microbiome in Health and Disease

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Abstract

Mammals are complex assemblages of mammalian and bacterial cells organized into functional organs, tissues, and cellular communities. Human biology can no longer concern itself only with human cells: Microbiomes at different body sites and functional metagenomics must be considered part of systems biology. The emergence of metagenomics has resulted in the generation of vast data sets of microbial genes and pathways present in different body habitats. The profound differences between microbiomes in various body sites reveal how metagenomes contribute to tissue and organ function. As next-generation DNA-sequencing methods provide whole-metagenome data in addition to gene-expression profiling, metaproteomics, and metabonomics, differences in microbial composition and function are being linked to health and disease states in different organs and tissues. Global parameters of microbial communities may provide valuable information regarding human health status and disease predisposition. More detailed knowledge of the human microbiome will yield next-generation diagnostics and therapeutics for various acute, chronic, localized, and systemic human diseases.

ABOUT THE HUMAN MICROBIOME

Metagenomics: identification of taxa by use of sequence-based culture-independent methods

Microbiome: a collection of microbial genomes

Microbiota: a collection of microbes

Cell-rich bacterial communities outnumber human cells in each person by an estimated ratio of 10 bacterial cells to each human cell (1). In other words, approximately 90% of the cells in and on the human body are microbial cells. To understand human metagenomics, the Human Microbiome Project was launched as part of the National Institutes of Health RoadMap version 1.5 in 2007 (2). Microbial diversity varies by anatomic site, and the complexity and aggregate functions of bacterial communities may correlate with an individual's health status, genotype, diet, and hygiene (3, 4). The numbers of different microbes (richness) at a body site, and the genetic diversity of microbiomes (5), are regulated partly by the local environment and biology of each body habitat. Although yeasts and viruses also form part of the human microbiome, the majority of published studies have focused on host-associated bacterial communities. Therefore, this review mainly discusses bacterial communities in the context of human anatomy and disease states. With respect to viruses and the human virome, diverse bacteriophage populations in the human microbiome are an additional source of biological diversity in human- and animal-associated bacterial communities (6).

Each human individual houses diverse microbial communities that reside in different body habitats, and these microbiomes differ greatly in terms of composition and function. The majority of human-associated bacteria fall within four phyla, Actinobacteria, Firmicutes, Proteobacteria, and Bacteroidetes (4, 7–9); each phylum contains many different bacterial species. The distribution and ratios of phyla differ with respect to body site (10). Body site–specific communities of bacteria vary to such a degree that the communities at each site (e.g., intestine) are more similar across the human population than they are to communities present at other sites within the same individual (4). The implication of these findings is that anatomical sites and tissues coevolved with microbiomes that contain different functional repertoires. Whereas the predominant microbial phyla and classes have been described, the microbiome is a fluctuating collection of genes and gene products; environmental perturbations, such as antibiotic treatment and infection, can readily alter the microbial composition and function of each community (9, 11). The human metagenome is relatively plastic or malleable, which makes the microbiome an attractive target for manipulation by cell or gene therapy.

The first year of human life provides an attractive window of opportunity to alter the composition and function of the microbiota in infants. Colonization of the newborn begins during the birth process, and relatively complex bacterial communities have been documented by the end of the first week of life (12, 13). The complexity of the bacterial community increases during infancy so that an adult-like complexity is attained by the end of the first year of life, and large fluctuations of bacterial populations occur during this first year (12, 13). Specific microorganisms within the community depend, in part, on environmental factors, including family size, nutrition, and water quality (3). The importance of diet was recently revealed in a study comparing the intestinal microbiomes of children from a village in the West African country of Burkina Faso with those of children in Florence, Italy. Although two bacterial phyla, Bacteroidetes and Firmicutes, dominated the microbiota of the population in each environment, there was a dramatic shift in terms of the relative percentages of these two dominant phyla: 73% and 12%, respectively, in Burkina Faso to 27% and 51%, respectively, in Italy (**Figure 1**) (14). Major differences in diet presumably shaped the microbiomes in these two pediatric populations in fundamental ways, possibly contributing to differences in whole-body metabolism (e.g., differences in fecal quantities of short-chain fatty acids) and development of organs and tissues.

The human microbiome may be a vast, malleable genome that can be modified by dietary, medical, and hygienic practices. Antibiotics

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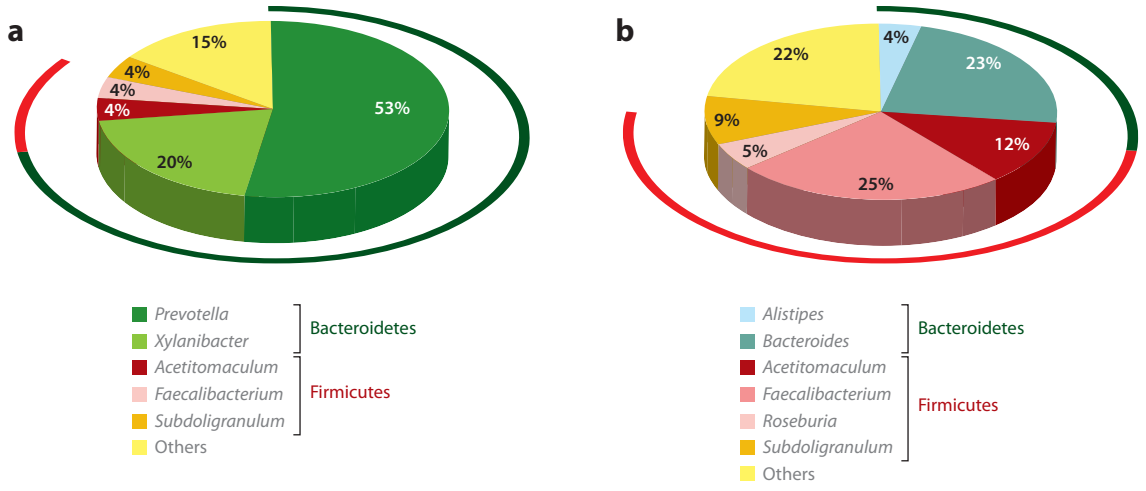


Figure 1

Bacterial taxa of the intestinal microbiota differ depending on diet. Taxa identified using 16S ribosomal RNA sequencing of DNA from fecal samples of children from (*a*) Burkina Faso and (*b*) Italy. The colors indicate differential distribution of classes of bacteria, including Firmicutes (*red*) and Bacteroidetes (*green*). Figure reproduced from Reference 14.

have potent effects by suppressing up to one-third of the gut microbiome with a simple five-day course of a single antimicrobial agent, ciprofloxacin. Gut communities are dynamic in nature such that most microbes returned to baseline levels within weeks posttreatment, but several bacterial taxa remained undetectable (11, 15). Such differences in microbial composition that arise from diet or medication history may contribute to different patterns of human disease predisposition and to the prevalence of various systemic and organ-specific disorders. Changes in human microbial populations have been linked to localized diseases [e.g., dental caries, bacterial vaginosis (BV)], as well as systemic disorders (e.g., autoimmune diseases) (9, 16). Alterations of the host environment (e.g., pH or nutrition) result in shifts in the relative abundance of pathogenic bacteria or upregulation of virulence genes in opportunistic pathogens, leading to disease states (7, 17, 18).

The yield of bacteriologic culture is limited because only 20%–60% of the microbes identified in different body sites have been cultured (2). Uncultured microbes can be identified by 16S ribosomal RNA gene sequencing (16S metagenomics), and entire genomes and metabolic pathways may be reconstructed by whole-genome shotgun (WGS) metagenomics (19). The reference genomes initiative within the Human Microbiome Project has facilitated identification of novel microbes and microbial genes (20), and such ongoing efforts will expand microbial sequence databases for microorganism identification and gene/functional annotation efforts. In addition to microbial DNA/RNA sequencing, metaproteomics and metabonomics strategies will enable investigators to identify microbial biomarkers that may be directly tied to differences in metabolic or physiologic functions in mammals. Databases that include microbial proteins and metabolites and newer bioinformatic tools that describe biochemical pathways in various bacteria (e.g., KEGG BRITE, U.S. Department of Energy Integrated Microbial Genomes) are providing new opportunities for human systems biology inclusive of metagenomics.

Taxon: phylogenetic classification of an identified or unidentified organism (plural: taxa)

Pathogen: an organism that is detrimental to the host or causes disease

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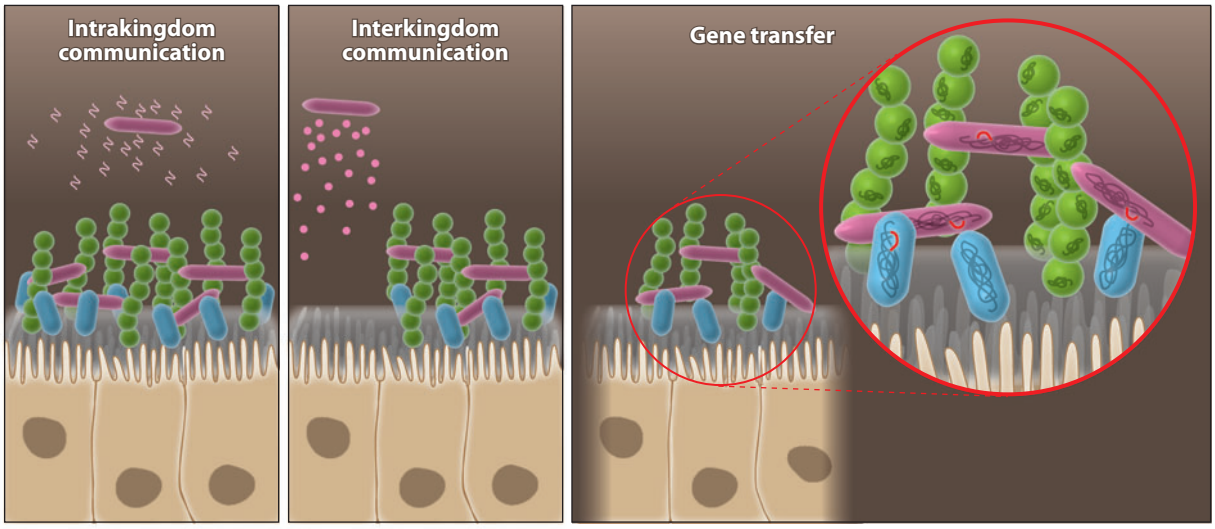


Figure 2

Basic modes of interspecies communication. Small molecules secreted by bacteria are used to communicate with other bacteria or the host, which results in the regulation of virulence factors or bacterial community composition, the regulation of gene expression in the host, or the supplementation of nutrients in the community as a whole. Communication can also occur via genetic exchange, in which a gene or genes, represented by the red region of DNA in the chromosome, are transferred between one species of bacteria and another. Genes involved in such transfer include genes involved in antibiotic resistance, virulence, and metabolism (26, 27, 31, 34, 44, 111–114).

INTERMICROBIAL COMMUNICATION AND GENE TRANSFER

Bacteria are social microorganisms that communicate and interact with one another as well as with the mammalian host (**Figure 2**). Individual bacterial species use small molecules to assess numbers of "self" (intraspecies communication) and to determine whether other bacterial species are present in the community (interspecies communication) by a mechanism termed quorum sensing (21). Signaling through quorum sensing enables group-specific behaviors, which cause changes in bacterial gene expression; some of these regulatory changes are potentially beneficial or detrimental to the host (22, 23). Probiotics can benefit the host by modulating quorum-sensing pathways in pathogenic bacteria. Such changes in quorum-sensing pathways by different species of *Lactobacillus* may result in the inhibition of toxin production (24–26). Quorum sensing–independent bacterial communication may result in the production of nutrients (e.g., vitamins) utilized by the human host and fellow microbes. Bacteria that produce cobalamin (or vitamin B_{12}) and folate can supplement the human host and nutrient-dependent bacteria in the community (27–30). In addition to producing vitamins and nutrients, various bacterial species modulate mucosal signaling pathways, resulting in changes in host gene expression and immune cell responses (**Figure 3**) (31). These data highlight the need for more research in bacteria–host interactions that will allow us to develop refined microbiome-derived diagnostics and therapeutics.

Quorum sensing: chemical signaling by microbes via secreted molecules

Probiotic: an organism that elicits health benefits to the host

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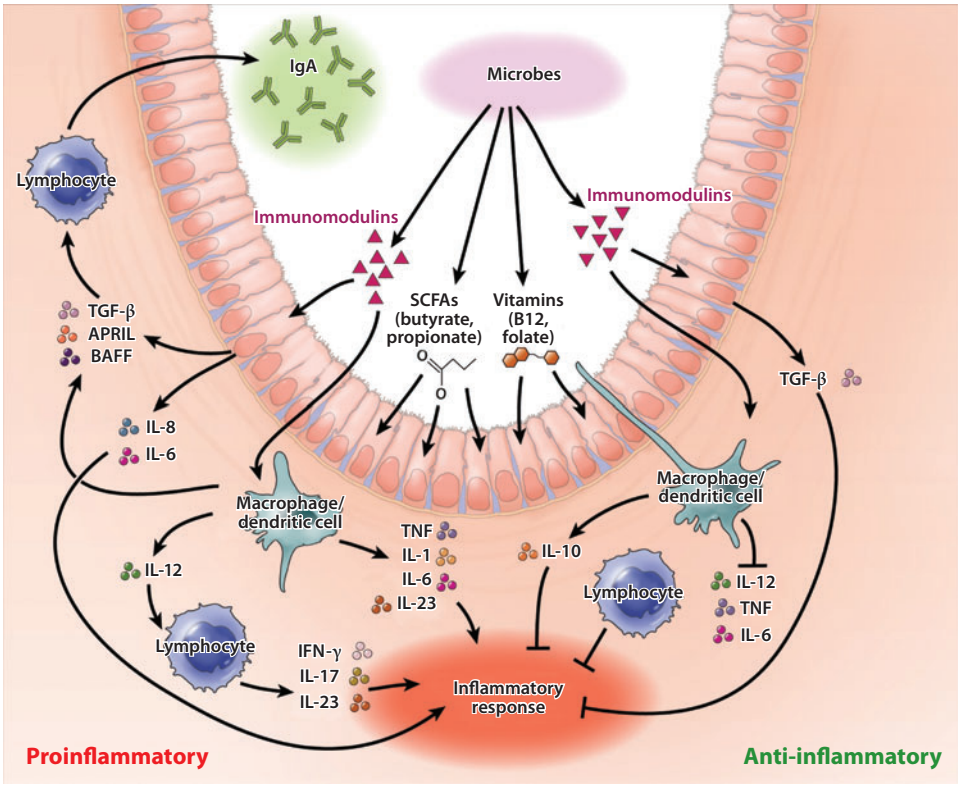


Figure 3

Intestinal bacteria produce nutrients and molecules that modulate mucosal immunity. Microbe-derived immunomodulins, short-chain fatty acids (SCFAs), and vitamins modulate host signaling, which leads to changes in cytokine and immune cell activity. Abbreviations: BAFF, B cell–activating factor; IFN, interferon; IgA, immunoglobulin A; IL, interleukin; TGF, transforming growth factor; TNF, tumor necrosis factor. Figure reproduced from Reference 115.

In addition to bacterial signaling via small molecules and nutrients, interspecies communication may include the lateral movement of genes between bacterial species and strains in microbial communities. Studies of gene transfer and acquisition within the oral cavity indicate that 5%–45% of genes present in bacteria with sequenced genomes are acquired through gene transfer (18). Gene transfer may be bidirectional such that virulence genes may be transferred from pathogens to commensals, and commensals may also serve as reservoirs of genes that encode antibiotic resistance and other genes that may facilitate digestion (**Figure 2**) (30, 32, 33). A recent study (34) described the transfer of gene(s) involved in carbohydrate utilization from bacteria present on seaweed to the gut microbiota of Japanese consumers. Presumably, this additional capability in Japanese gut microbiomes enhanced the ability of these individuals to digest and absorb dietary seaweed and associated algal carbohydrates. Spatial relationships between microbes facilitate genetic exchange and signaling among microbes within these complex communities.

ORAL MICROBIOME: INSIGHTS INTO DENTAL CARIES AND PERIODONTAL DISEASE

The oral cavity contains a diverse set of niches, including the soft tissues of the tongue and

Gene transfer: transfer of a novel genetic element from one organism to another

Commensal: a colonizing organism that is neither beneficial nor detrimental to the host

Biofilm: a three-dimensional bacterial community encased within an exopolysaccharide

tonsils, saliva, and the hard surfaces of the teeth. Despite this range in habitats, a similar array of bacteria constitutes the oral microbiome in each niche of a healthy oral cavity (35–37). This core microbiome is consistent throughout the human population, as the extent of interpopulation variation of the microbiome is similar to the extent of intrapopulation variation (38). The formation of biofilms on the supragingival surface of teeth constitutes what is commonly termed dental plaque (18, 39). Such biofilms consist predominantly of various combinations of streptococcal species (~60% of the bacteria in dental plaque), and metabolism of carbohydrates by streptococci and other bacteria results in degradation of tooth enamel via pH reduction at the tooth surface (18, 39). This reduction in pH is partially attributable to organic acid production by streptococci and other lactic acid bacteria. Differences between the microbiota of healthy and diseased oral cavities of children in China and the United States were recently assessed in culture-independent studies (40). Species of the genus *Granulicatella* were more abundant in the plaque samples of caries-prone children from the United States and less abundant in a similar population from China (40, 41). The data suggest that changes in the microbiota of dental plaque predispose susceptible individuals to dental caries, and ethnic differences may account for the impact of specific components of the microbiome. This concept is consistent with the so-called ecological plaque hypothesis, which states that changes in the microbiota from a healthy state shape the environment in a way that leads to dental plaque and decay (42).

Inflammatory lesions within the subgingival crevice and degradation of periodontal tissue associated with subgingival biofilms are characteristics of periodontal disease (43, 44). Several bacterial pathogens, including *Porphyromonas gingivalis* and *Aggregatibacter actinomycetemcomitans*, are associated with periodontal disease. These bacteria interact with epithelial cells, which leads to an alteration of the epithelial cell transcriptional profile, including bacteria-specific changes in the apoptotic and cell cycle–progression pathways (44, 45). Infection of human immortalized gingival keratinocytes (HIGKs) with the oral pathogen *P. gingivalis* causes the induction of HIGK genes involved in cell-cycle regulation, specifically cyclin-dependent kinases (44). Because the subgingival crevice is not colonized solely by *P. gingivalis*, the pathogen is found in communities with various commensal microbes, including several streptococcal species, and the collective regulation imposed upon host cells by these communities of bacteria may differ significantly from that observed by a single species. Studies of gene-expression profiles of HIGKs infected with cocultures of *P. gingivalis* and *Streptococcus gordonii* found that *S. gordonii* significantly modulated host cell responses to *P. gingivalis*, including the downregulation of genes involved in cell-cycle regulation (44).

Differences between the microbiota present in the gingival crevice of healthy and diseased tissue from 49 individuals were recently investigated (43). The data indicated that several species of oral streptococci, as well as *Veillonella parvula*, were associated with the healthy state. Although no major differences in the ratios of bacterial species were observed between healthy and disease states, greater numbers of bacteria were associated with periodontal lesions (43). Overgrowth of multispecies bacterial communities at specific infection sites, rather than changes in ratios of bacterial species, may drive the pathogenesis of periodontal disease.

MICROBIOMES AND METABOLISM: DIABETES AND OBESITY

The human diet plays a role in shaping the composition of the human microbiome, and the microbiome, in turn, affects the ability of the individual to absorb and metabolize nutrients. Differences in gut microbial composition between lean and obese individuals stimulated ample interest in commensal microbes and in how the human microbiome may be relevant to human health and disease. In fundamental studies using mouse models, obese mice contained a relative abundance of Firmicutes, in contrast

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to the relative preponderance of the phylum Bacteroidetes in lean mice (46, 47). The gut microbiomes of lean mice, when transplanted into *ob/ob* (obese) mice, normalized the body weights of the latter, indicating that differences in microbial composition could affect body metabolism and energy harvest and influence the predisposition of mammals to obesity. Mice deficient in the microbial pattern-recognition receptor Toll-like receptor 5 (TLR5) displayed hyperphagia, became obese, and developed features of the metabolic syndrome, including hypertension, hypercholesterolemia, and insulin resistance (48). When gut microbiomes from these mice were transplanted into germ-free mice with an intact *TLR5* gene, recipient mice developed similar features of the metabolic syndrome, which suggests that the intestinal microbiome, and not murine TLR genetics, was the key determinant of this disease phenotype.

Studies in mouse models were correlated with similar findings in human individuals and stimulated interest in how the gut microbiome may affect predisposition to metabolic disorders. Long-term dietary trends shape the intestinal microbiota and metabolic activity of the host. In a study that compared shifts in the gut microbiota of lean and obese but otherwise healthy men in the United States, a 20% increase in bacteria of the phylum Firmicutes was observed in association with increased caloric absorption and energy harvest by bomb calorimetry (49). Prior data suggested that Fiaf (fasting-induced adipocyte factor) is a contributing factor to enhanced fat deposition by mammals with a conventional gut microbiome (50). The suppression of this gut epithelium–derived lipoprotein lipase inhibitor is essential for gut microbiota–induced deposition of triglycerides in adipocytes. Recent WGS metagenomics data uncovered the predominance of three basic enterotypes or human gut microbiome profiles in a group of 39 individuals representing six nationalities (51). Through the mining of whole-metagenome data, two microbial ATPase complexes were identified as potential biomarkers of the microbiome that correlate strongly with body mass index in humans; such biomarkers support the potential link between the capacity of the gut microbiome for energy harvest and obesity. In other words, an enhanced ability to harvest energy may be associated with a tendency toward fat deposition and obesity in individuals.

spp.: several species

An individual's predisposition to harvest and store fat may be determined in utero because changes in the mother's gut microbiome may translate into alterations of the intestinal microbiomes of their infants; the vertical transmission of microbiomes associated with increased energy harvest may result in infants with inherited tendencies to excessive weight gain. A recent study by Collado et al. (52) highlighted the abundance of *Clostridium histolyticum*, *C. difficile*, and *Akkermansia* spp. in overweight mothers and their infants. The transition from development in utero to infancy emphasizes the potential role of human nutrition, combined with human development, in shaping the human microbiome early in life. Dynamic fluctuations in the human microbiome have been detected at both macroscopic and microscopic levels, and diet and mammalian development influence the composition and function of the human microbiome.

The gut microbiome can affect whole-body metabolism and alter physiological parameters in multiple body compartments (53). Gut microbial communities fundamentally alter the metabolite composition of body fluids, the liver, and the kidneys. In one study (53), gnotobiotic mice had increased quantities of phosphocholine and glycine in the liver and increased quantities of bile acids in the intestine. The gut microbiome also influences kidney homeostasis by modulating quantities of key cell-volume regulators such as betaine and choline (53). A more recent study showed specific differences in the patterns of bile acids and overall bile acid diversity in germ-free versus conventional rats (54). Germ-free animals contain greater quantities of conjugated bile acids in the heart and liver and greater quantities of taurine, compared with conventional animals. Because bile acids are now recognized as important cell-signaling molecules, as evidenced

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by effects on farnesoid X receptor–regulated pathways, such differences in microbial bile acid cometabolite patterns may have important consequences for whole-body metabolism.

Intestinal microbiomes have also been studied in the context of insulin resistance in adult patients with type 2 diabetes. In one study using relatively deep tag-encoded sequencing (55), the relative abundance of the phylum Firmicutes and the class Clostridia was significantly reduced in adults with type 2 diabetes (55). Additionally, the ratios of Bacteroidetes to Firmicutes and *Bacteroides–Prevotella* to *C. coccoides–Eubacterium rectale* groups were correlated with plasma glucose concentrations in adult patients. Such findings are intriguing and prompt questions regarding how microbial composition and corresponding metabolites may influence whole-body metabolism in humans. Intestinal bacterial taxa differ with respect to their abilities to utilize dietary carbohydrates and host-derived carbohydrates (e.g., mucus components) (47, 56). Bacteroides species contain a rich collection of carbohydrate-utilization pathways, and such gut bacteria can easily assimilate dietary carbohydrates. However, in situations of dietary carbohydrate starvation, the gut bacteria turn to mucus in the gastrointestinal tract as a source of carbohydrate, thereby compromising the mucus barrier. Such fluctuations in diet may have functional consequences for bacteria and the host so that the "cannibalization" of indigenous mammalian carbohydrates may result in predisposition to invasion or inflammation. Insights into the metabolic pathways of indigenous microbes with respect to carbohydrate utilization and metabolism may provide an explanation of the mechanisms of differential energy harvest and endocrine/metabolic disorders in humans.

THE MICROBIOME AND GASTROINTESTINAL DISEASES

The Esophagus and Stomach

Until recently, the esophagus and stomach were considered to contain relatively simple microbial communities, with the occasional presence of specific pathogens. Advances in metagenomics resulted in widespread appreciation of the presence of rich microbial communities in these body habitats, and shifts in these populations may be associated with an increased risk of disorders of chronic inflammation and neoplasia. With respect to the lower esophagus, two basic types of esophageal microbiomes were associated with different physiological states (57). The type I microbiome was dominated by the genus *Streptococcus*, and this community class was found mostly in individuals lacking evidence of esophageal disease. The type II microbiome was characterized by greater phylogenetic diversity, including various gram-negative anaerobic and microaerobic bacteria. The type II microbiome was correlated with esophagitis and Barrett's esophagus in one group of patients. This report included a relatively superficial data set of only 200 sequenced clones per sample, and unsupervised cluster analysis of these limited data sets yielded two basic microbiome types in the esophagus (57). Deeper sequencing studies will be needed to ascertain the significance of these findings, but the ability to stratify patients on the basis of classification of human microbiomes is an important observation. The association of increased microbial diversity and human disease phenotypes may hinge on the relative complexity of the microbiome at a specific body site. Body sites that have coevolved with relatively simple microbial communities such as the esophagus may not be capable of sustaining more complex communities in the absence of disease.

The discovery of *Helicobacter pylori* in the human stomach by Warren & Marshall (58) in the 1980s fueled interest in bacterial colonization and infection of the human stomach. Subsequent studies and clinical trials with antimicrobial agents confirmed the importance of *H. pylori* in atrophic gastritis, peptic ulcer disease, and susceptibilities to gastric MALT (mucosa-associated lymphoid tissue) lymphoma and gastric adenocarcinoma (59). Following a period of exuberance regarding the

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apparent success of antimicrobial therapy in patients with peptic ulcer disease, Blaser & Falkow (3) emphasized the potential danger of elimination of one long-time component of the human gastric microbiome. Deficiencies of *H. pylori* in modern human communities have been associated with an increased risk of Barrett's esophagus and esophageal adenocarcinoma in these populations. Studies of the gastric microbiome documented the presence of multispecies microbial communities in the human stomach, indicating that many species, in addition to gastric *Helicobacter*, apparently colonize the stomach (15, 60). The removal of *H. pylori* from the gastric microbiome may reduce the risk of peptic ulcer disease and gastric adenocarcinoma, but conversely, the absence of this species may cause changes to the esophageal microbiome that increase the risk of chronic esophageal inflammation and cancer. The effects in the esophagus may arise from alterations in the nature of esophageal microbiomes (type I or type II) due to the presence or absence of *H. pylori*.

Acute Gastroenteritis and Infectious Colitis

Acute gastroenteritis includes diarrheal diseases caused by various infectious agents, including viruses, bacteria, and protozoal pathogens. Diagnosis of the causative agent of acute gastroenteritis may defy routine methods in microbiology. Viral agents are more predominant in children younger than three years of age, with a shift to a predominance of bacterial pathogens in children older than three years of age (61). In one study, only 47% of stool samples that underwent complete diagnostic testing yielded a specific etiologic agent (61). The pathobiology of acute gastroenteritis includes effacement of intestinal villi, enhanced intestinal permeability due to interactions between pathogens and the gut epithelium, and toxin production that mediates disruption of the intestinal barrier and immune cell infiltration.

Studies in mouse models of acute gastroenteritis suggest that microbial richness may be markedly diminished in the gastrointestinal tract during episodes of disease, and such a compromised microbiome may enhance disease susceptibility and pathogenesis. Reduced diversity may mean that specific components that protect the host from pathogenic invaders are absent. In a mouse model of hemorrhagic colitis arising from *Escherichia coli* 0157:H7, bifidobacterial species protect the host by providing genes that encode specific ATP-binding cassette-type carbohydrate transporters (62). Protection by bifidobacteria appears to be mediated, at least in part, by the production of acetate and by acetate's ability to inhibit Shiga toxin translocation across the intestinal epithelium. The commensal bacteria in the intestine may inhibit the ability of toxins and virulence factors to penetrate the mucosa, thereby preventing acute infection and leading to mucosal pathology.

Antimicrobial therapy, especially specific classes of agents such as β-lactams and fluoroquinolones, predisposes subsets of patients to antibiotic-associated diarrhea and colitis due to specific pathogens. Treatment with antibiotics, including single-agent therapy, may yield profound effects on the composition and function of the gastrointestinal microbiome. A five-day course with a single fluoroquinolone resulted in the elimination or suppression of approximately one-third of the fecal microbiome within three to four days of antibiotic treatment (11). Gut microbial populations mostly recovered within one week of completion of a course of antibiotics, but the recovery of the microbiome was incomplete (15). Similarly, diminished bacterial diversity in the respiratory tract due to antimicrobial therapy was associated with ventilator-associated pneumonia during the treatment course (63). Antimicrobial-associated disorders, part of a new class of disorders of microbial ecology, result from the short- and long-term impact of antibiotics on the composition and function of the human microbiome.

Toxigenic *C. difficile* is the primary etiologic agent of antimicrobial-associated diarrhea (AAD), and it accounts for an estimated 15%–25% of cases. Other clostridial etiologies include enterotoxin-producing strains of

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Dysbiosis: loss of balance within an animal-associated microbial community

C. perfringens and possibly *C. spiroforme*, a species known to cause disease in rabbits, which inhabits the human gastrointestinal tract. Enterotoxin-producing strains of *Staphylococcus aureus* have been implicated as another possible microbial etiology of AAD (64). Clearly, perturbations of the gastrointestinal microbiota create opportunities for multiple bacterial species to proliferate and cause disease. Disease may be secondary to production of sufficient amounts of toxin such that a threshold is passed, resulting in symptomatic diarrheal illness. Reduced diversity of the intestinal microbiome is associated with recurrent *C. difficile*–associated disease in adults (65). Patients with restricted phylogenetic diversity appear to be predisposed to recurrent disease, which supports the role of the intestinal microbiome as a protective barrier to colonic infection. Sufficiently diverse microbial communities are presumably able to effectively suppress the proliferation of pathogens and subsequent production of enterotoxins by pathogens such as toxigenic *C. difficile* in the intestine.

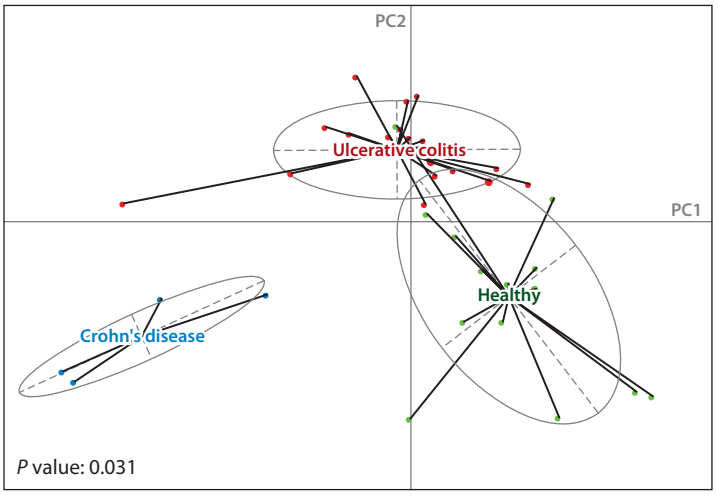


Figure 4

Inflammatory bowel disease states are correlated with microbiome composition. Principal components analysis of bacterial species representing ≥1% of the microbiome of fecal samples from healthy subjects ($n = 14$), patients with ulcerative colitis (21 subjects) or Crohn's disease (4 subjects). First components 1 and 2 (PC1 and PC2) are plotted on the x and y axes, respectively. Figure reproduced from Reference 75.

Fecal transplantation is a microbiome-based strategy to restore microbial balance and species richness in patients with recurrent *C. difficile* disease (66, 67). Greater phylogenetic diversity is associated with reduced risk of *C. difficile*–associated disease, and fecal transplantation via colonoscopy demonstrated that diverse donor microbial communities can supplant disease-associated microbiomes (68). Such introduction of phylogenetic diversity in a diseased human individual is associated with clinical recovery and amelioration of symptoms. Instead of microbiome transplantation, the simultaneous addition of specific beneficial microbes to treatment regimens offers new opportunities to promote microbial diversity and resilience of the host. A series of systematic reviews reported evidence recommending probiotic preparations to treat *C. difficile*–associated diarrhea in children or adults (69) or to prevent antibiotic-associated diarrhea in children (70).

Inflammatory Bowel Disease

Inflammatory bowel disease (IBD), including Crohn disease and ulcerative colitis, probably has multifactorial etiologies including human gene- and microbiome-associated components. Mutations in human genes such as *NOD2* affect the recognition of microbial patterns or signals, and *NOD2* alleles are associated with Crohn disease in a subset of individuals (71–73). In addition to human genetic defects, microbial dysbiosis has been implicated in inflammatory bowel disease (74). In one study, a set of 155 bacterial species of the fecal microbiome separately clustered patients with Crohn disease, patients with ulcerative colitis, and healthy controls into different groups by principal components analysis (**Figure 4**) (75); a prior study had shown that patients with IBD can be effectively distinguished from healthy controls on the basis of qualitative interpretation of global microbiome data (76). The segregation of patient populations according to bacterial DNA sequences in the human microbiome without human DNA-sequence information highlights the potential importance of metagenomics in

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human pathology. In addition to differences between patients with IBD and healthy individuals, different disease phenotypes of Crohn disease are associated with differences in gut bacterial populations (77). Human metagenomics may lead to new diagnostic strategies that will refine disease stratification in combination with histopathologic assessment of IBD.

The bacterium *Faecalibacterium prausnitzii* is a member of the *C. leptum* group commonly found in the fecal microbiomes of healthy adult individuals (78), and deficiency of *F. prausnitzii* appears to be relatively specific for ileal Crohn disease (79). Deficiencies of *F. prausnitzii* are associated with increased frequency in endoscopic recurrence of active Crohn disease in adult patients (80). Intraperitoneal injection of soluble components of *F. prausnitzii* into mice subsequently exposed to trinitrobenzene sulfonate, a chemical agent that induces a lethal acute colitis, protected mice from intestinal injury and mortality. The protective effects in vivo were associated with anti-inflammatory activities through the use of human intestinal epithelial cell culture models, and they appear to be due to small, unidentified organic compounds, or immunomodulins, produced by gut bacteria. Deficiencies of Firmicutes such as *F. prausnitzii* may determine susceptibility to Crohn disease (81) and suggest new directions for microbiome-targeted therapies (82). Interestingly, intraperitoneal inoculation of this organism is associated with remote effects such as the modulation of eight or more metabolites in the urinary compartment alone. The presence of *F. prausnitzii* in humans causes the modulation of eight urinary metabolites of diverse structure and stresses the potential multicompartment effects of gut bacteria (82). In terms of diagnostic utility, bacterial features such as the amounts of *F. prausnitzii* in fecal specimens have been used to diagnose active Crohn disease and ulcerative colitis (81). The presence of *F. prausnitzii* and other commensals in fecal specimens is also inversely correlated with appendicitis, which indicates that certain bacteria may suppress inflammation in susceptible individuals (83). In the same study, the presence of *Fusobacterium* spp. correlated positively with the severity of appendicitis, and the bacteria were visualized in cecal biopsy specimens by fluorescence in situ hybridization (**Figure 5**) (83).

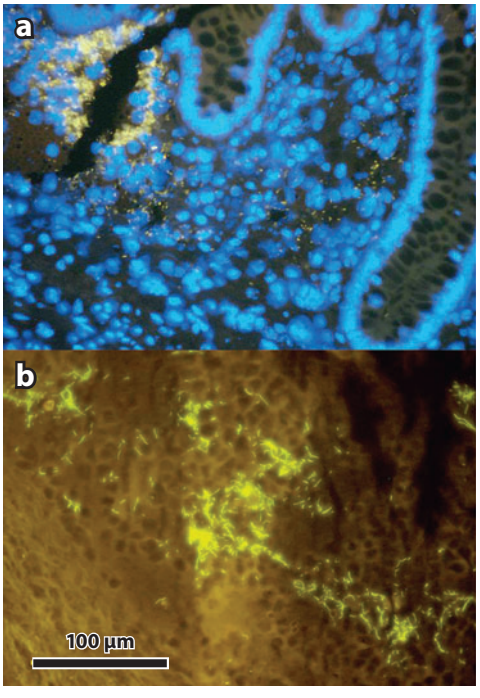


Figure 5

Bacterial abundance correlates with the severity of appendicitis, as demonstrated by fluorescence in situ hybridization. (*a*) *Fusobacterium necrophorum* and (*b*) *F. nucleatum* associated with suppurative appendicitis. Figure reproduced from Reference 83.

Patients with IBD and other diseases of the intestine may undergo surgical procedures to address refractory disease phenotypes. Patients who have undergone small-bowel transplantation and postsurgical ileostomy procedures undergo dramatic transformations of the human microbiome, and such changes in composition within the ileal fluid are associated with environmental factors such as oxygen concentration (84). The local ambient atmosphere may affect the ileal fluid directly, allowing facultative anaerobes to dominate the intestinal microbiota. Obligate anaerobes of the *Clostridium* and *Bacteroides* genera have been

detected in ileal samples immediately following ileostomy closure, which suggests that restoration of anaerobiosis in the intestine follows this procedure. Restoration of the microbiome may represent an important component of future improvements in intestinal rehabilitation and transplantation outcomes following abdominal surgery.

Recurrent Abdominal Pain and Irritable Bowel Syndrome

Recurrent abdominal pain (RAP) is widely prevalent in children and adults and accounts for approximately 30% of health care visits for children aged 4–16 years (85). RAP includes two primary disorders known as functional abdominal pain (pain only) and irritable bowel syndrome (IBS; pain plus diarrhea or constipation). Clinical trials with probiotics, or beneficial bacteria, in adults and children provided evidence that microbial dysbiosis somehow contributes to the disease phenotype of chronic abdominal pain and that differences in gut microbial communities may contribute to differences in pain signaling and nociception (86). In another study (87), germ-free mice demonstrated increased motor activity and reduced anxiety early in life, and these changes in behavior were correlated with changes in gene expression in the brain. These studies indicate that the gut microbiome influences early brain development in mice with effects on mammalian behavior (87) and that the addition of lactobacilli may affect visceral pain sensitivity in rodents (88). Fluctuations in the gut microbiome may affect pain signaling in the enteric nervous system and visceral pain perception.

In addition to differences in the gut microbiomes of patients with IBS, subtypes of IBS may also be distinguished by differences in gut microbiome composition. IBS with diarrhea is associated with reductions in *Lactobacillus* spp., and IBS with constipation is associated with elevated abundance of *Veillonella* spp. (89). A separate study also found increased abundance of *Veillonella* spp. in adult patients with IBS, and this difference was associated with increased production of acetate and propionate in these individuals (90). The first 16S metagenomics study of adults with IBS documented enrichment of the phyla Proteobacteria and Firmicutes (especially Lachnospiraceae) in these patients (91). Published results with Proteobacteria, especially γ-Proteobacteria, are consistent with our own findings (91a), demonstrating greater proportions of these bacteria in children with IBS. Specific bacteria, notably taxa of the genus *Alistipes*, are more abundant in the gut microbiomes of children with moderate-to-severe abdominal pain (91a). Patients who had IBS with diarrhea yielded diminished quantities of the genera *Bacteroides* and *Bifidobacterium* (91). The latter finding is intriguing because *Bifidobacterium* species, and not *Lactobacillus* species, were effective in diminishing the symptoms of IBS in one clinical trial (92). Differences in the intestinal microbiome may be exploited to refine strategies for microbial manipulation therapies and nutritional management of patients with RAP. Metagenomics-based diagnostic tests may be developed to refine the ability of the pathologist and gastroenterologist to define subtypes of IBS that can be effectively managed.

VAGINAL AND URETHRAL MICROBIOMES: DISEASE IMPLICATIONS

The vaginal microbiome in women has implications for pregnancy and preterm birth, sexually transmitted diseases (STDs), and conditions such as vaginitis and BV. In healthy women of reproductive age, the genus *Lactobacillus* predominates; it includes species such as *L. iners*, *L. crispatus*, *L. gasseri*, and *L. jensenii* (93). Although these four *Lactobacillus* species dominate the composition of the vaginal microbiome, the relative proportions of different bacterial species vary with ethnicity and vaginal pH. Interestingly, one group of women had a phylogenetically diverse microbiome with

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different anaerobic bacteria and a relative deficiency of lactobacilli (93), and women with different vaginal microbiomes may differ with respect to predisposition to STDs or BV. To understand how maternal microbiomes influence neonatal microbiomes, a vertical transmission study described the phenomenon in which the composition of oral and gut microbiomes in neonates depends on the mode of delivery at birth. Infants delivered vaginally acquire bacterial communities in their skin, oral cavity, and nares that resemble their mothers' vaginal microbiota, dominated by *Lactobacillus*, *Prevotella*, and *Sneathia* spp. (94). Infants delivered by Cesarean section harbor microbial communities similar to those found on human skin, dominated by *Staphylococcus*, *Corynebacterium*, and *Propionibacterium* spp. How babies are born may partly determine the composition of bacterial communities in different body habitats early in life, and these differences may have implications for early susceptibility to disease (e.g., sepsis, necrotizing enterocolitis).

In contrast to studies implicating reductions of microbial diversity associated with different disease states (e.g., disorders of the gastrointestinal tract), studies of BV suggest that increased bacterial diversity may be associated with disease. As in the esophagus, body sites with restricted microbial diversity in the healthy state may not tolerate increased microbial diversity without adverse consequences. Through the use of next-generation sequencing in a population of Chinese women, increased phylogenetic diversity and the presence of many low-abundance bacterial taxa have been associated with BV (**Figure 6**) (95). This study was consistent with prior publications (96) indicating that increased bacterial species richness and diversity correlate with the disease phenotype of BV. *L. iners* is the predominant species in BV-negative women but is markedly reduced in BV-positive women (95). Genera such as *Gardnerella*, *Sneathia*, and *Megasphaera* have been detected at higher prevalence and relatively greater abundance in BV-positive women. Next-generation DNA pyrosequencing found a number of low-abundance bacterial genera that had not previously been detected, and particular species may be useful microbial biomarkers for disease. The genus *Atopobium* is present in 84% of women diagnosed with BV, in contrast to 22% of women in the control group. The absence or relative deficiency of indigenous, nonpathogenic lactobacilli in the vagina may predispose women to recurrent urinary tract infections (97), and urinary tract pathogens may displace indigenous lactobacilli by the production of natural antibiotics such as bacteriocins (98). Dysbiosis of the vagina apparently predisposes women to various disorders, such as BV, and recurrent infections. More recent 16S metagenomics data of vaginal microbiomes indicate that the microbial composition of the vagina fundamentally differs between pregnant and nonpregnant women (K. Aagaard, K. Riehle, T.A. Mistretta, J. Ma, C. Coarfa, C. Huttenhower, D. Gevers, S. Rosenbaum, I. Van den Veyver, A. Milosavljevic, J. Petrosino & J. Versalovic, manuscript submitted). Such differences may yield insights into the vaginal microbiome, predisposition to preterm birth, and susceptibility to infections in pregnancy.

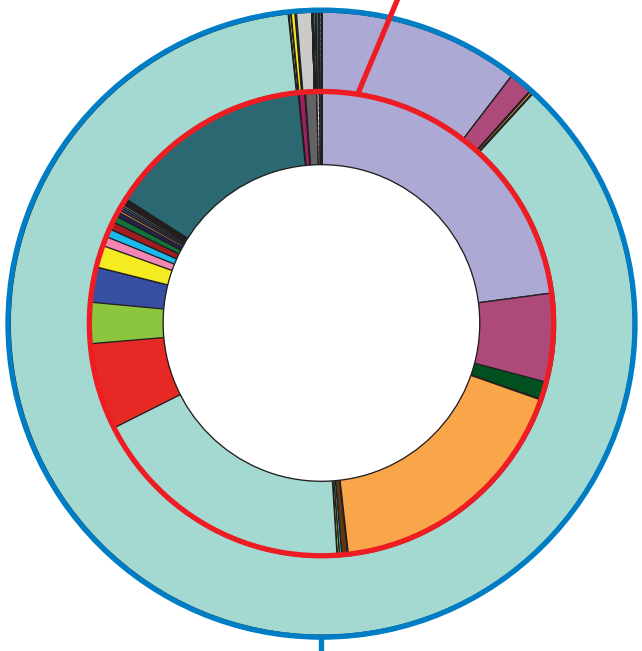
The urethra and surrounding skin serve as primary sites for genitourinary tract–associated microbial communities in men. The male urinary microbiome differs between men without evidence of infection and men with evidence of asymptomatic sexually transmitted infection (STI) (99). Men with STIs caused by the bacteria *Chlamydia*, *Neisseria*, *Mycoplasma*, or *Ureaplasma* have urinary microbiomes with diverse, fastidious, and uncultured bacteria that are rare in STI-negative men. Phylogenetic clustering methods such as Unifrac clearly separated STI-positive and STI-negative men on the basis of principal components analysis. Uncultured bacteria associated with pathology of the female genital tract are abundant in urine specimens of STI-positive men, and these findings suggest that the male urethra contains pathogenic bacteria that cause infections when inoculated into the female vagina.

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Other BV-positive genera identified (relative abundance < 0.25%)

Olsenella
Arcanobacterium
Slackia
Mobiluncus
Bifidobacterium
Cryptobacterium
Micromonospora
Tessaracoccus
Corynebacterium
Couchioplanes
Okibacterium
Propioniferax
Bacteroides
Porphyromonas
Tannerella
Kaistella
Owenweeksia
Fluviicola
Xylanibacter
Crocinitomix
Flavobacterium
Flectobacillus
Mariniflexile
Persicivirga
Capnocytophaga
Cloacibacterium
Dysgonomonas
Winogradskyella
Peptoniphilus
Pseudobutyrivibrio
Lachnospiraceae genera incertae sedis
Alloiococcus
Veillonella
Anaerococcus
Coprococcus
Subdoligranulum
Anaeroglobus
Fastidiosipila
Turicibacter
Abiotrophia
Moryella
Pediococcus
Propionispira
Bulleidia
Catonella
Butyrivibrio
Gracilibacter
Lactovum
Paralactobacillus
Pelospora
Mitsuokella
Paucisalibacillus
Roseburia
Sporobacter
Acetivibrio
Acidaminococcus
Dolosigranulum
Finegoldia
Granulicatella
Caloranaerobacter
Cerasibacillus
Desulfonispora
Eremococcus
Isobaculum
Marinilactibacillus
Anaerobacter
Anaerobaculum
Brevibacillus
Faecalibacterium
Lachnobacterium
Atopococcus
Atopostipes
Two Clostridiaceae genera incertae sedis
Ignavigranum
Sporacetigenium
Staphylococcus
Tissierella
Ureibacillus
Vagococcus
Propionigenium
Streptobacillus
Cetobacterium
Leptotrichia
Acinetobacter
Ureaplasma
TM7 genera incertae sedis
Caldilinea
Cycloclasticus
Haemophilus
Sutterella
Serratia
Campylobacter
Erythrobacter
Neisseria
Comamonas
Hylemonella
Sphingomonas
Sphingopyxis

BV positive



Gardnerella
Atopobium
Eggerthella
Prevotella
Hallella
Lactobacillus
Shuttleworthia
Megasphaera
Papillibacter
Streptococcus
Dialister
Gemella
Aerococcus
Parvimonas
Peptostreptococcus
Alloiococcus
Sneathia
Fusobacterium
Mycoplasma

BV negative

Other BV-negative genera identified (relative abundance < 0.25%)

Eggerthella
Olsenella
Arcanobacterium
Slackia
Mobiluncus
Bifidobacterium
Cryptobacterium
Micromonospora
Tessaracoccus
Corynebacterium
Couchioplanes
Okibacterium
Propioniferax
Prevotella
Hallella
Bacteroides
Porphyromonas
Tannerella
Kaistella
Owenweeksia
Fluviicola
Xylanibacter
Crocinitomix
Flavobacterium
Flectobacillus
Mariniflexile
Persicivirga
Capnocytophaga
Cloacibacterium
Dysgonomonas
Winogradskyella
Shuttleworthia
Megasphaera
Papillibacter
Streptococcus
Dialister
Gemella
Aerococcus
Parvimonas
Peptostreptococcus
Peptoniphilus
Pseudobutyrivibrio
Lachnospiraceae genera incertae sedis
Veillonella
Anaerococcus
Coprococcus
Subdoligranulum
Anaeroglobus
Fastidiosipila
Turicibacter
Abiotrophia
Moryella
Pediococcus
Propionispira
Bulleidia
Catonella
Butyrivibrio
Gracilibacter
Lactovum
Paralactobacillus
Pelospora
Mitsuokella
Paucisalibacillus
Roseburia
Sporobacter
Acetivibrio
Acidaminococcus
Dolosigranulum
Finegoldia
Granulicatella
Caloranaerobacter
Cerasibacillus
Desulfonispora
Eremococcus
Isobaculum
Marinilactibacillus
Anaerobacter
Anaerobaculum
Brevibacillus
Faecalibacterium
Lachnobacterium
Atopococcus
Atopostipes
Two Clostridiaceae genera incertae sedis
Ignavigranum
Sporacetigenium
Staphylococcus
Tissierella
Ureibacillus
Vagococcus
Sneathia
Fusobacterium
Propionigenium
Streptobacillus
Cetobacterium
Leptotrichia
Acinetobacter
Mycoplasma
Ureaplasma
TM7 genera incertae sedis
Caldilinea
Cycloclasticus
Haemophilus
Sutterella
Serratia
Campylobacter
Erythrobacter
Neisseria
Comamonas
Hylemonella
Sphingomonas
Sphingopyxis

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SKIN MICROBIOME AND DERMATOLOGIC DISEASES

Many factors, including climate, occupation, and hygiene, shape the human skin microbiome, and intrinsic factors such as physiology, genotype, or disease state (100) and skin microbiomes provide unique information about individuals that may be used in forensic pathology (101). The microbiota of the skin are characterized by the same four predominant phyla as in other body sites: Actinobacteria, Firmicutes, Proteobacteria, and Bacteroidetes (4, 102, 103). However, the relative distributions of bacterial phyla and families differ significantly among skin sites throughout the human body (103, 104). Human skin at different body sites varies in terms of temperature, humidity, glandular distribution, and environmental exposure (103). Areas of the body that are enriched with sebaceous glands have prominent populations of Actinobacteria (e.g., Corynebacterineae, Propionbacterineae), in contrast to relatively dry areas, such as the volar forearm, that are enriched for Proteobacteria. Despite bacterial diversity at different sites, human skin–associated microbial composition is more similar between skin sites than among other body habitats (103). Interestingly, when bacteria characteristic of the tongue or forearm were transplanted to the forehead, the characteristic forehead bacteria were able to outcompete the bacteria from the tongue or forearm (4), which highlights the importance of the biology of the body habitat in shaping the composition and function of microbiomes. Even when the skin surface is compromised, as in a diabetic wound, bacterial populations on the skin may have a profound impact on the ability of wounds to heal, and they reepithelialize the skin surface by unknown mechanisms (105).

Skin colonization by pathogens such as *S. aureus* is a prerequisite for subsequent *S. aureus* infection of the skin and other body sites. The nasal microbiome contains *S. aureus* in approximately 50% of the adult population, and the composition of the microbiome may predispose individuals to bacterial infections (106). The nasal microbiomes of hospitalized patients are deficient in Actinobacteria, especially *Propionibacterium acnes*, and these bacterial deficiencies are inversely correlated with the relative abundance of staphylococcal species in inpatients (106). Persistent infections have been associated with changes in the skin microbiota, such as diminished abundance of the genus *Propionibacterium* and the phylum Actinobacteria (106). Additionally, the relative abundance of *S. aureus* is inversely correlated with the abundance of the commensal *S. epidermidis*. These shifts or differences in microbiomes at skin surfaces may yield important prognostic information related to risk of infection in vulnerable patient populations. *S. epidermidis* is a common skin commensal that can participate in remodeling microbial communities by production of antibacterial peptides (107). Relative distributions of bacteria may have implications for the relative abundance of drug-resistant bacteria. Nasal carriage of methicillin-resistant *S. aureus* (MRSA) has been associated with MRSA infection, and the relative preponderance of drug-resistant bacteria may depend on skin microbiome composition.

Dermatologic disorders are beginning to be linked to changes in the skin microbiome. In addition to infections of the skin and other body sites, the presence of *S. aureus* has been

Figure 6

Bacterial vaginosis (BV) is associated with a different and more complex vaginal microbiome. This figure depicts the relative abundance of bacteria identified by 16S ribosomal RNA from vaginal swabs of 50 healthy (BV-negative) subjects and 50 BV-positive subjects. Figure reproduced from Reference 95.

associated with persistent dermatologic disorders such as atopic dermatitis (107). However, more information is needed to determine the relative contribution of *S. aureus* versus other bacteria in triggering atopic dermatitis. Fluctuations in skin bacterial populations have been associated with psoriasis, a chronic inflammatory disorder that may be induced by pathogens such as *S. aureus* and *S. pyogenes* (102). Targeted polymerase chain reaction strategies highlight differences in the distribution of specific bacterial genera at skin sites, and these approaches may be useful for monitoring disease progression and the management of psoriasis (102). In addition to changes in bacterial communities, differences in fungal communities may contribute to skin disorders such as seborrheic dermatitis and tinea versicolor. The fungal genus *Malassezia* is the most widely prevalent fungal taxon on human skin, and *Malassezia* species are differentially distributed (108, 109). Knowledge regarding fluctuations of specific yeasts may be useful for disease prevention and management in dermatology.

POTENTIAL MECHANISMS

The human microbiome has emerged as a core component of human systems biology and genomics, and serious consideration of differences and fluctuations in the human microbiome will be important for one to fully understand basic mechanisms of human pathology. Global features such as excessive bacterial numbers or diversity at specific body sites may contribute to inflammation and pathologic host responses. Conversely, reduced richness of bacteria at body sites such as the intestine may diminish the ability of individuals to resist infection, assimilate nutrients, or maintain aggregate function of a healthy microbiome. In addition to these global features, specific differences or fluctuations of bacterial species may cause enhanced predisposition to disease. The combinations of human genotypes and microbiome types (e.g., enterotypes) may cause increased predisposition to disease and increased risk of recurrent chronic diseases. The site of pathology may depend partly on the nature of microbiomes at specific body sites when combined with local patterns of gene expression and epigenomics. The presence or relative abundance of specific bacteria may protect individuals from disease phenotypes through the production of signals or compounds that counteract abnormalities of human physiology. The timing and spatial development of microbiomes early in life may have lasting consequences on differentiation and maturation of different mucosal surfaces, organs, and tissues. In addition to local effects, microbiomes contribute to whole-body metabolism and may have remote effects on human physiology.

Once the specific composition of the microbial community and interactions within the microbiota are identified, this information can be utilized to manipulate microbial communities either by antibiotics, diet, or applications of specific biologicals or by supplementation with natural or engineered microbes (**Figure 7**) (110). By performing microbial manipulation with antibiotics, probiotics, or dietary interventions, microbiomes may be shifted or remodeled such that these microbial communities could help tilt the balance from a diseased state to a healthy state (**Figure 8**).

SUMMARY AND FUTURE DIRECTIONS

The conceptual framework of human disease must accommodate the composition and function of human-associated microbiomes. Differences among and fluctuations in human microbiomes in various body habitats will provide key insights into mechanisms of disease, and such findings will result in next-generation diagnostics and therapeutics. Prior to developing applications in human medicine, investigators must acquire more mechanistic information regarding the biology of the human microbiome and its relevance to pathology.

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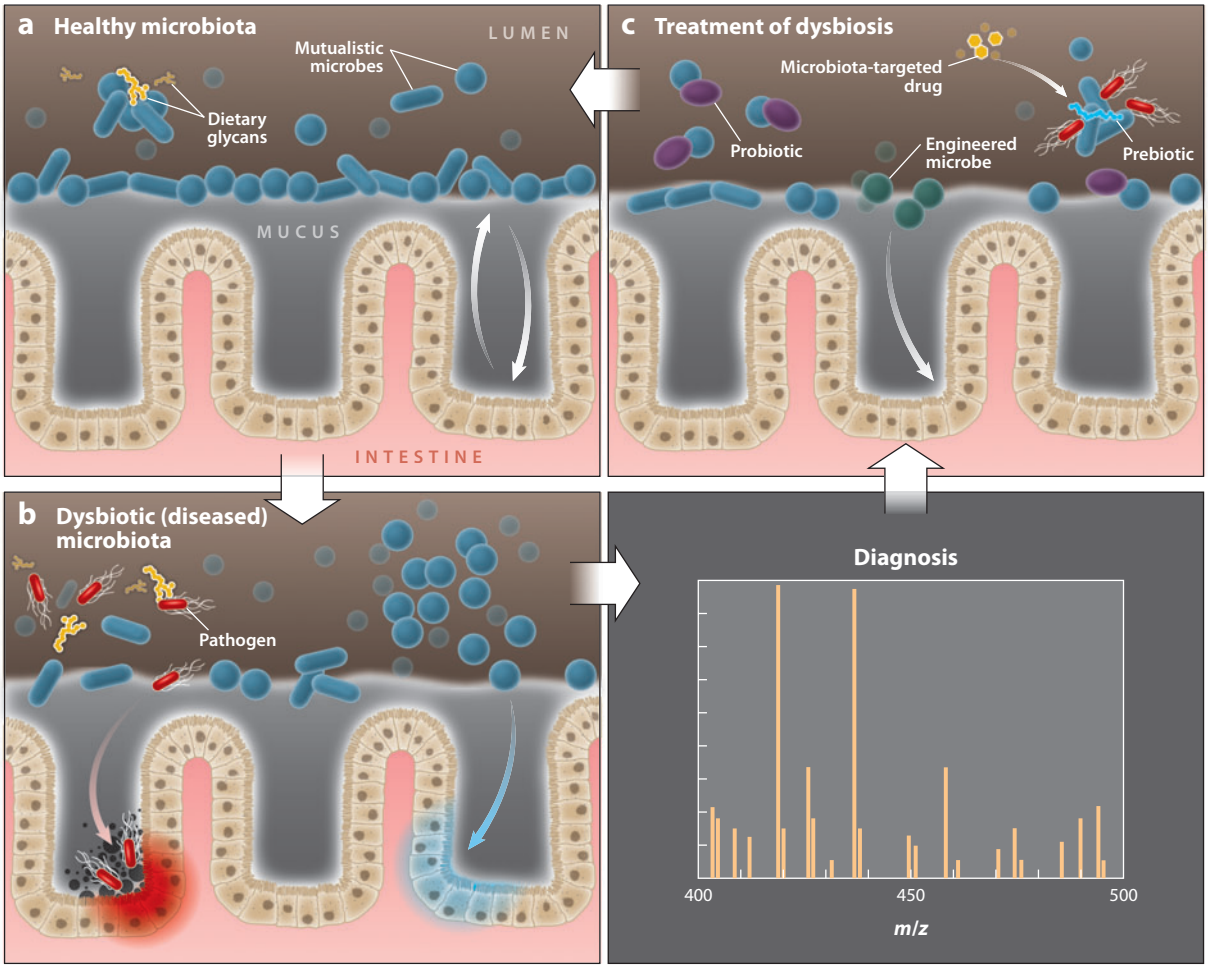


Figure 7

Microbial manipulation and treatment of dysbiosis. Differences found between (*a*) healthy and (*b*) diseased microbiota may lead to factors that can be exploited (*c*) to treat dysbiosis. Figure reproduced from Reference 110.

Specific microbes or bacterial species distributions are still being defined in healthy cohorts and compared with those of patients with various disease phenotypes. A paucity of mechanisms, coupled with many intriguing findings, highlights the urgency of this field of biomedical research. Recent publications suggest that identification of specific microbes may be a sensitive barometer of temporal fluctuations in human disease because the variation in microorganisms vastly exceeds the extent of variation in known metabolic genes and pathways. For applications such as forensic pathology, 16S metagenomics or organism identification may be more useful. By contrast, functional and WGS metagenomics, with a focus on specific pathways and mechanisms of disease, will probably be a more robust approach for most diagnostic and therapeutic applications.

Applications of metagenomics may include the incorporation of microbial sequencing data

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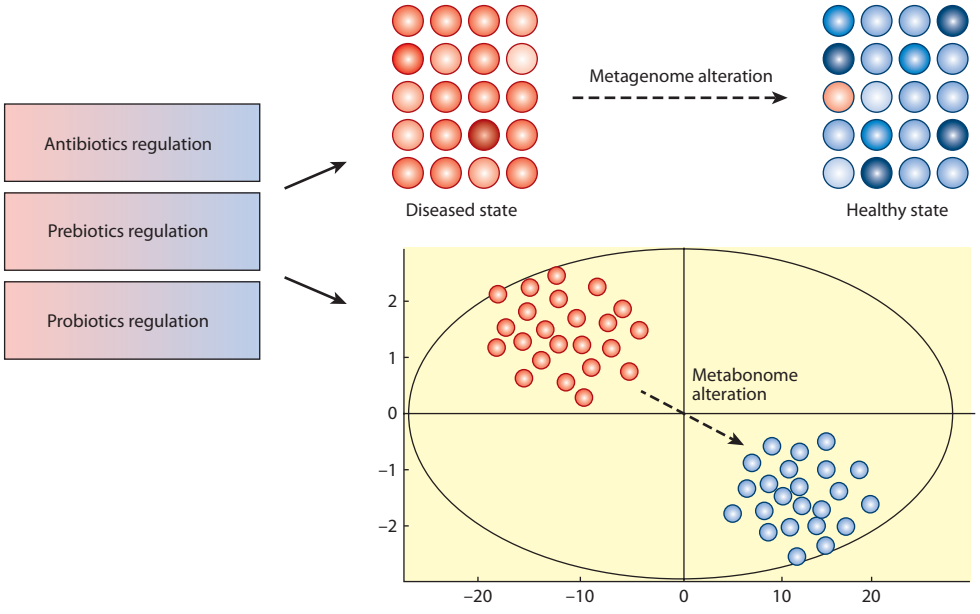


Figure 8

Restoring the metabonome to a healthy state by microbial manipulation. Alterations of the composition of the microbiota and subsequent metabonomic profile by microbial manipulation with antibiotics, probiotics, or nutritional interventions that may include prebiotics. Figure reproduced from Reference 116.

in the diagnostic workup of chronic immune-mediated and inflammatory diseases that previously relied only on histopathologic or immunologic assessment. In addition to microbial DNA and RNA targets, microbial biomarkers such as proteins and metabolites may emerge as useful diagnostic features and indicators of disease progression versus successful disease management. For some disorders, convenient specimens such as oral or skin swabs and self-collected stool may be sufficient to provide relevant information in the diagnostic laboratory. In other cases, deeper body sampling such as endoscopy or surgery may be required to obtain specimens with the requisite microbial genes and biologics for clinical evaluation. The remote effects of microbiomes in distant or multiple body compartments must also be considered in future investigations and applications of human microbiome research. The remote effects of the human microbiome may be especially important when considering the prognosis and management of systemic disorders. For example, metabonomics of urinary specimens may yield a robust signal-to-noise ratio when exploring distant effects of the gut microbiome on whole-body metabolism.

In addition to development of new diagnostic strategies, next-generation probiotics and microbe-derived biotherapeutics based on advances in compositional and functional metagenomics may be important for future management of gastrointestinal, skin, oral, and other disorders. Intentional manipulation of the human microbiome may facilitate the recovery of patients by effects on diverse physiologic parameters such as pain signaling, systemic immune responses, energy harvest, and whole-body metabolism.

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SUMMARY POINTS

1. The human microbiome is an integral component of the human body.
2. The human microbiome represents a plastic metagenome that varies according to body site, environmental exposure, and health status within and between individuals.
3. Pathologic alterations, or dysbioses, of site-specific bacterial communities can shift microbiomes from healthy states to disease-associated states.
4. Bacteria produce signals and compounds that affect local microbial populations and local tissue compartments.
5. Human-associated bacterial communities may affect multiple remote body compartments and may yield systemic effects on energy harvest, immunity, and whole-body metabolism.
6. Increased diversity of the microbiome may be associated with disease at body sites that normally have restricted diversity. Conversely, reduced diversity of the microbiome may be associated with disease at body sites with greater indices of diversity.
7. Microbial manipulation strategies, including human nutrition, antibiotics, and microbial supplementation (probiotics), may provide new strategies for disease management and prevention.

FUTURE ISSUES

1. A greater understanding of the signaling mechanisms of the human microbiome, including intra- and interkingdom interactions, may result in a refined approach to human systems biology.
2. Explorations of global features of the microbiome, such as richness, evenness, diversity, and how they influence disease predisposition, may be useful for disease management.
3. Investigations of specific features and discriminant taxa of the microbiome provide opportunities for important discoveries of microbial genes and microorganisms. Such classes of microbes may affect diagnosis and treatment strategies.
4. Metagenomics and an improved understanding of the dynamism of microbiomes at different body sites can be translated into rational manipulations by diet, probiotics, and new drug combinations.

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DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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